

Digital Transformation

The use of technology to radically improve performance or reach of an enterprise

GLOSSARY 01 - SHEET 1 / 8

PROFESSIONAL DEFINITION

Digital transformation is the use of digital technology to radically improve the performance or reach of an enterprise, achieved by combining investment in technology (digital intensity) with the leadership capability required to drive change through the organisation (transformation-management intensity). It is distinguished from technology deployment by the requirement that operating model, customer experience and business processes are redesigned, not merely automated (Westerman, Bonnet and McAfee, 2014; Vial, 2019).

WHO DEFINED IT ORIGINALLY

The working definition used in this course is Westerman, Bonnet and McAfee's (2014) formulation, developed at the MIT Center for Digital Business with Capgemini Consulting from a study of 391 large firms. Vial (2019) later consolidated 282 works into an integrative process definition that frames transformation as a process in which digital technologies trigger structural change and generate value. Earlier antecedents include Venkatraman's (1994) IT-enabled business transformation levels.

WHY THIS DEFINITION WAS DEVELOPED

Firms were spending heavily on technology and reporting no financial improvement. The MIT-Capgemini team needed a construct that separated firms buying technology from firms changing how they work, in order to explain why investment and performance were uncorrelated. Defining transformation as a two-dimensional capability - technology plus governance - made the difference measurable and gave boards a diagnostic rather than a slogan.

HOW IT IS USED

Score the firm on digital intensity and on transformation-management intensity, place it in one of four quadrants, and use the gap between the two scores to decide whether the next euro should be spent on technology or on governance.

WHEN IT IS USED

At the opening of a transformation programme, at annual strategy review, when a portfolio of pilots has failed to reach production, and whenever a new leadership team inherits a technology agenda it did not set.

BY WHOM IT IS USED

Chief executives and boards setting the transformation mandate, chief digital and information officers owning delivery, and business unit heads accountable for the operating model that must actually change.

FIGURE 1 - THE TWO INTENSITIES OF TRANSFORMATION

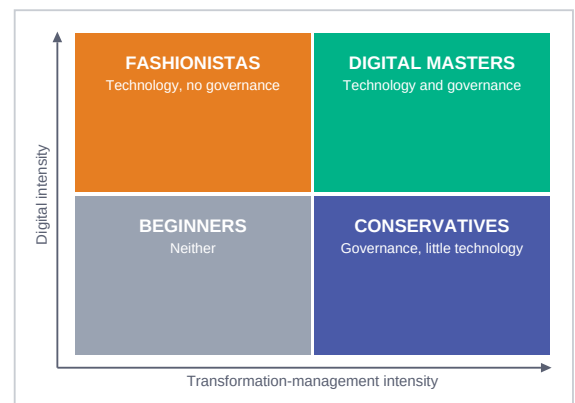


Figure 1. Digital transformation as the joint product of two intensities. Only the upper-right quadrant is associated with superior revenue generation and profitability; the Fashionista quadrant is the characteristic failure mode of technology-led programmes (after Westerman, Bonnet and McAfee, 2014).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor construct was IT-enabled business transformation (Venkatraman, 1994), which ranked change on a single ladder from localised exploitation to business scope redefinition. Westerman and colleagues replaced the ladder with two orthogonal axes, which allowed the empirically important failure case - heavy technology investment with weak governance, the Fashionista - to be named and observed. Vial's (2019) process definition then added the feedback loop: transformation changes the environment that triggered it.

RELATED DEFINITIONS IN THIS COURSE

Digital Maturity - Unit 1

Digitisation vs Digitalisation - Unit 1

Business Model Innovation - Unit 1

AI Adoption - Unit 4

AI Strategy - Unit 3

SOURCES (HARVARD REFERENCING STYLE)

- Venkatraman, N. (1994) 'IT-enabled business transformation: from automation to business scope redefinition', Sloan Management Review, 35(2), pp. 73-87.
- Vial, G. (2019) 'Understanding digital transformation: a review and a research agenda', Journal of Strategic Information Systems, 28(2), pp. 118-144.
- Westerman, G., Bonnet, D. and McAfee, A. (2014) Leading Digital: Turning Technology into Business Transformation. Boston, MA: Harvard Business Review Press.
- Westerman, G., Calmejsane, C., Bonnet, D., Ferraris, P. and McAfee, A. (2011) Digital Transformation: A Roadmap for Billion-Dollar Organizations. Cambridge, MA: MIT Center for Digital Business and Capgemini Consulting.